

FULL CYBERSECURITY SUMMARY REPORT

ICT Department — Weekly Cybersecurity Practical Assignment

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1. Introduction

This report summarizes all cybersecurity practical activities conducted from Day 1 to Day 5. The assessment focused on evaluating the institution's network security, application security, password security, phishing awareness, firewall configurations, and overall cybersecurity posture.

The practical exercises involved:

- Network vulnerability scanning
- Password security auditing
- Email security analysis
- Firewall and access control review
- Security recommendations and awareness development

The objective was to identify security weaknesses and recommend improvements to strengthen the institution's cybersecurity environment.

2. Summary of Activities Performed

Day	Activity
Day 1	Network Infrastructure Vulnerability Scan
Day 2	Password Security Audit & Policy Review
Day 3	Phishing Awareness & Email Security Check
Day 4	Firewall & Network Access Control Review
Day 5	Full Security Summary & Recommendations

3. Day 1 — Network Infrastructure Vulnerability Scan

Practical Work Performed

Network discovery and scanning were conducted using Kali Linux and Nmap.

Commands Used

```
nmap -sn 192.168.0.0/24
nmap -sV 192.168.0.1
nmap -sV 192.168.0.110
```

Findings

Devices Identified

- Tenda Router
- TP-Link Device
- Dell Workstation
- Intel-based Device
- Multiple unknown devices

Open Services Detected

Device	Open Port	Service
Router	80/tcp	HTTP Management
Workstation	3389/tcp	Remote Desktop Protocol

Risks Identified

- Router management accessible through HTTP
- Open Remote Desktop service
- Unknown devices connected to the network
- Lack of visible network segmentation

Recommendations

- Enable HTTPS router management
- Restrict RDP access
- Segment administrative and employee systems
- Implement stronger Wi-Fi controls

4. Day 2 — Password Security Audit & Policy Review

Practical Work Performed

Password exposure checks were conducted using:

- [Have I Been Pwned](#)

Email Checked

academicperfect@gmail.com

Findings

The email address was found in multiple known data breaches involving:

- Stolen passwords
- Credential stuffing datasets
- Stealer malware logs

Risks Identified

- Password reuse risk
- Compromised credentials
- Unauthorized account access

Recommendations

- Use strong unique passwords
- Enable Multi-Factor Authentication (MFA)
- Change compromised passwords immediately
- Use password managers

5. Day 3 — Phishing Awareness & Email Security Check

Practical Work Performed

Email security records were analyzed using DNS queries.

Commands Used

```
dig TXT contextexpertagencieslimited.com
dig TXT _dmarc.contextexpertagencieslimited.com
dig TXT default._domainkey.contextexpertagencieslimited.com
```

Findings

Security Record	Status
SPF	Configured
DKIM	Configured
DMARC	Missing

Risks Identified

- Increased phishing exposure
- Email spoofing possibility
- Lack of DMARC enforcement

Recommendations

- Configure DMARC policies
- Conduct phishing awareness training
- Enable spam filtering
- Monitor suspicious email activities

6. Day 4 — Firewall & Network Access Control Review

Practical Work Performed

Firewall and access controls were assessed through network scans and service analysis.

Findings

Device	Observation
Router	HTTP management enabled
Workstation	RDP port exposed
Other Devices	Filtered ports detected

Security Strengths

- Some firewall filtering already active
- Internal devices partially protected

Weaknesses

- Open management services
- Unknown devices connected
- Limited segmentation

Recommendations

- Restrict administrative access
- Enable firewall logging
- Use MAC filtering
- Implement WPA3 wireless security

7. Top 5 Security Risks Identified

Risk	Description
1. Missing DMARC	Increased phishing and spoofing risk
2. Open RDP Service	Potential remote access attacks
3. HTTP Router Management	Unencrypted administration access
4. Compromised Email Exposure	Password reuse and credential theft
5. Unknown Network Devices	Possible unauthorized access

8. Overall Recommendations

Immediate Actions

1. Implement DMARC on all domains
2. Enable Multi-Factor Authentication (MFA)
3. Restrict RDP access
4. Upgrade wireless security to WPA3

5. Remove unknown devices from the network

Long-Term Improvements

- Conduct regular vulnerability scans
- Implement centralized cybersecurity monitoring
- Introduce employee cybersecurity awareness training
- Deploy enterprise endpoint protection systems
- Perform regular security audits

9. Lessons Learned

During the cybersecurity practical assignments, the following skills and concepts were learned:

- Network vulnerability assessment
- Port and service scanning
- Password breach analysis
- Email authentication verification
- Firewall evaluation techniques
- Security reporting and documentation
- Cybersecurity risk prioritization

10. Conclusion

The cybersecurity assessment revealed that the institution has implemented some security protections such as SPF, DKIM, and partial firewall filtering. However, several important risks remain unresolved, including missing DMARC protection, exposed remote services, and insufficient network segmentation.

Addressing these weaknesses through stronger authentication, firewall hardening, phishing protection, and continuous monitoring will significantly improve the organization's overall cybersecurity posture.

11. Tools Used

- Kali Linux
- Nmap
- Dig DNS Utility
- Wireshark
- DNS Email Security Analysis
- [Have I Been Pwned](#)
- [GRC ShieldsUP!](#)